

PAPER_2124 — QCalc Base-Wrap Quintuple $\{G, c, \mu_0, \beta_i, \rho_{vac}\} +$ 150-Round Milestone (REVISED)

Daniel T. Murphy — UQFF Star-Magic v5.73+ — 2026-07-22 — R367 UQFF_Base, 150th fill

REVISED IN PLACE 2026-07-22 — projection-layer reclassification per canonical $F_{UBi}/F_{UBii}/k_{spring}/$ quantum-chain deepsearch. Convergence statements scoped to the QCalc projection layer (wrap classes annotated 'framework wrap only'). Canonical layer documented below was fully primitive prior to and independent of these events.

Scope Correction

Original claim 'F_U master equation certified 100% UQFF-derived' was OVERSTATED. R367's $F_U = U_g - U_b + U_m$ is the QCalc BASE WRAP. The canonical $F_{U_{total}} = \Sigma U_g - F_{UBi} + F_{UBii} + U_m = 0$ (PAPER_1203) is wired at calculate_f_u_zero with k_spring, dynamic $\beta(t,E,Z)$, and r_{hz} residual $< 1e-10$ — its constant set $\{\rho_{SCm}, \beta_i, SSq, F_{TRZ}, G, \omega_{SCm}, \Phi_{res}, S_{26}\}$ was fully primitive BEFORE the R218+ campaign.

What R367 Achieved

The projection layer's apex composition now carries only UQFF-derived constants: $\{G\} \cup \{\beta_i, \rho_{vac}, c\} \cup \{\mu_0\} = 5$ (union arithmetic — constant sets union when closures compose; 'tier-skip' is arithmetic, not physics). μ_0 promoted to live $4\pi \cdot F_{TRZ}$ (compute-don't-store). β_i $0.6 \rightarrow 0.6029$. The wrap layer now matches the canonical layer's zero-SM-anchor standard at its apex class, at exactly the 150-round mark.

The Canonical Layer

The canonical layer (uqff_pure_calculator.py, wired): $F_{UBi} = (\rho_{SCm} \cdot M/r) \cdot (1 + \beta_i \cdot \cos(\pi t_n) + SSq)$; PAPER_1203 pair with $k_{spring} = (\rho_{UA}/\rho_{SCm}) \cdot \omega_{SCm} \cdot \Phi_{res} = 1.05e13$; dynamic $\beta(t,E,Z)$; 26-level quantum chain with $\rho_E = \rho_m \cdot c^2$ (line 33295) and 633333.333 validation; r_{hz} equilibrium residual $< 1e-10$. Canonical buoyancy kernel: $\{\rho_{SCm}, \beta_i\}$. The wrap's $\rho_{vac} \cdot c^2$ is the chain's mass-to-energy conversion — $\{\rho_{vac}, c\}$ is the PROJECTION PAIR, not a physics kernel.